

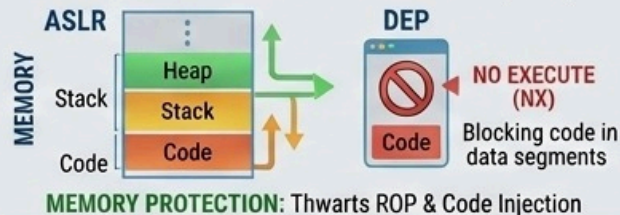
FEATURE HIGHLIGHT

MANDATORY ACCESS CONTROL (MAC)



FEATURE HIGHLIGHT

ADDRESS SPACE LAYOUT RANDOMIZATION (ASLR) & DATA EXECUTION PREVENTION (DEP)



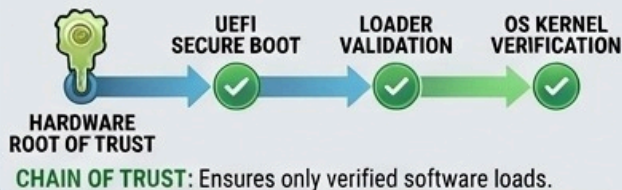
FEATURE HIGHLIGHT

KERNEL PATCH PROTECTION (KPP)



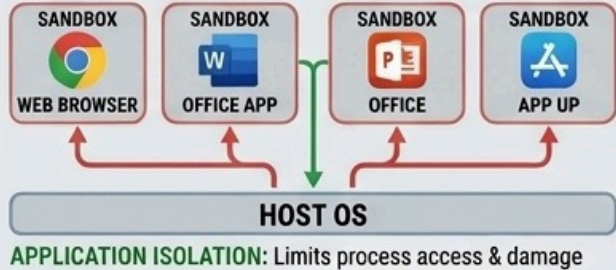
FEATURE HIGHLIGHT

TRUSTED BOOT CHAIN



FEATURE HIGHLIGHT

SANDBOXING & CONTAINERIZATION



THE SECURE KERNEL

SECURE INTER-PROCESS COMMUNICATION (SIPC)

FEATURE HIGHLIGHT

CRYPTOGRAPHY SYSTEMS

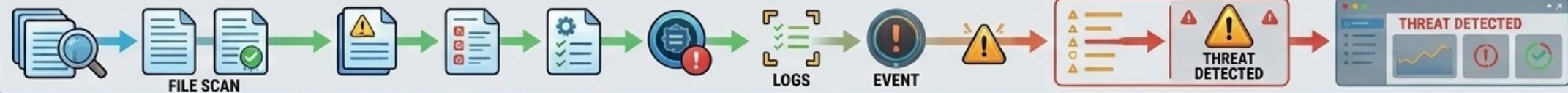


FEATURE HIGHLIGHT

CRYPTOGRAPHIC SERVICES



INTEGRATED SECURITY AUDITING & MONITORING



★ Key Features of a Secure OS



Trusted Boot Chain –

Cryptographically verified firmware and OS loader. Ensures integrity from power-on.



Mandatory Access Control (MAC) –

Enforces strict access policies based on object sensitivity. Prevents data leakage.



Application Sandboxing –

Isolates processes in their own environments. Restricts access to sensitive system resources.



Memory Protection (ASLR/DEP) –

Thwarts overflow attacks by randomizing address layouts. Prevents execution of injected code.



Network Security (Firewall/VPN) –

Full-state inspection and encrypted tunnels. Secures all data in-transit.



Secure Inter-Process Communication –

Verified and controlled message-passing between components. Prevents privilege escalation.



Data Encryption (At Rest) –

Comprehensive file and full-disk encryption. Keeps sensitive data private even if physically stolen.



Security Auditing & Monitoring –

Detailed logging of system events and access attempts. Facilitates threat detection.